

18.06 Linear Algebra - Lecture 10/12/9

Textbook Problem Sets

11-06-2026

Problem Set 3.4 pg 164

1) $Ax = 0$ $A = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$ (3×4)

$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{bmatrix} \Rightarrow R$

$Rx = 0$
 $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Set $x_4 = 1$:

$x_3 + (1) = 0$
 $x_3 = -1$

$x_2 - (1) = 0$
 $x_2 = 1$

$x_1 - (1) = 0$
 $x_1 = 1$

$x = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix} \therefore x = c \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$

2) $V = \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$

$\left. \begin{array}{l} 4 \text{ pivot variables} \\ 2 \text{ free variables} \end{array} \right\} \text{all six vectors in } \mathbb{R}^4$

$V_1, V_2, V_3 \rightarrow$ Independent. (-1 diff position)

$\hookrightarrow$ 3 pivots $\rightarrow \mathbb{R}^3$ subspace.

so any 4 or more vectors are dependent.
 largest independ # = 3.

5) a) vectors $(1, 3, 2)$; $(2, 1, 3)$; $(3, 2, 1)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -13 \\ 0 & -1 & -5 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -13 \\ 0 & 0 & 12 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -13 \\ 0 & 0 & 12 \end{bmatrix}$$

$\Rightarrow$ 3 pivot columns $\therefore$ Independent ✓

$$\begin{aligned} (-5) \cdot 5(-1) &= 0 \\ (-13) - 5(-5) &= 12 \end{aligned}$$

$\hookrightarrow$ $n \times n$ matrix

invert $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -7 \\ 0 & 0 & -18/5 \end{bmatrix}$: invertible $\Rightarrow$ independent columns

b) $\begin{bmatrix} 1 & 2 & -3 \\ -3 & 1 & 2 \\ 2 & -3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 2 & -3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 0 & -7 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -3 \\ 0 & 7 & -7 \\ 0 & 0 & 0 \end{bmatrix}$

$\hookrightarrow$ zero in pivot position $\rightarrow$ not invertible $\rightarrow$ dependent.

11) a) $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \rightarrow$ cols dependent $\rightarrow$ subspace is a line $c \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ✓

b) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$ Plan in $\mathbb{R}^3$ ✓

c) All of $\mathbb{R}^3$

d) All $\mathbb{R}^3$

13) a) $\left. \begin{aligned} \dim C(A) &= 2 \\ \dim C(U) &= 2 \\ \dim R(A) &= 2 \\ \dim R(U) &= 2 \end{aligned} \right\}$ ✓

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{pivots} = 2.$$

$\dim = 2$

$\hookrightarrow R(A) = R(U)$ $\left\{ \begin{array}{l} \text{rowspace } U \text{ combination of } A \text{ rows} \\ \text{rowspace } A \text{ combination of } U \text{ rows} \end{array} \right.$

rank = $\dim C(A) = \dim C(U) = \dim R(A) = \dim R(U)$

$\hookrightarrow$ Fundamental property of a matrix $\rightarrow$ same after elimination

15) If $v_1, \dots, v_n$ are linearly independent, the space they span has dimension n ✓.

These vectors are a basis ✓ for that space.

If vectors are the columns of an m by n matrix, the m is smaller than or equal to n . If $m = n$, that matrix is square.

↳ "not less" → $(m \geq n)$

↳ Invertible.

↳ Read as one statement → same n throughout.

18) ~~24~~ $v_1, \dots, v_6$ are six vectors in $\mathbb{R}^4$.

a) Those vectors might not ✓ span $\mathbb{R}^4$

b) Those vectors might be ~~x~~ are not linearly independent. $\exists$ x indep span $\mathbb{R}^4$ ✓

c) Any four of those vectors might be ✓ a basis for $\mathbb{R}^4$.

Read!!!

21) 5×5 matrix A basis for $\mathbb{R}^5$

a) Equation $Ax = 0$ only solution $x = 0$ because columns are indep ✓ all vectors are independent.

b) If b is in $\mathbb{R}^5$ then $Ax = b$ is solvable because basis vectors span $\mathbb{R}^5$ ✓ columns span.

↳ describe vectors in A as columns of A .

→ Note: basis → $\exists$ solution for every b .

24) ~~23~~ a) True → ~~x~~ False $A = \begin{bmatrix} 1 & 1 \end{bmatrix}$ → dep cols; indep rows.

b) True → ~~x~~ False Col Space $\neq$ Row space for $A = \begin{bmatrix} 0 & 0 \end{bmatrix}$

c) True → dimension of row and column space equal r , matrix rank

d) True → column space defined False → columns could be dependent on each other.

26) a) $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$ ✓

b) ~~$\left\{ \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right\}$~~

c) $\left\{ \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \right\}$

29) a) All of $\mathbb{R}^3$

b) Line in $\mathbb{R}^3$

c)

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ All of $\mathbb{R}^3$

Span space of all

~~all~~ $\times$ ~~all~~ (span)

$\hookrightarrow$ each rank

span space of all

$\hookrightarrow$ Span all m

31) a) All functions to satisfy $\frac{dy}{dx} = 0$.
 $\hookrightarrow y(x) = c$

b) Choose particular function to satisfy $\frac{dy}{dx} = 3$
 $y(x) = 3x$ ✓

c) Find all functions for $\frac{dy}{dx} = 3$;
 $y(x) = 3x + c$ ✓

34) $\dim = 1$:

$y_1(x) = 3x$

$y_2(x) = 6x$

$y_3 = 9x$

$\dim = 2$:

$y_1(x) = 3x$

$y_2(x) = x^2$

$y_3 = 6x$ ✓

$\dim = 3$:

$y_1(x) = 3x$

$y_2(x) = x^3$

$y_3(x) = x$

$\hookrightarrow$ other examples

26) a) $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$ ✓

b) ~~$\left\{ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\}$~~

c) $\left\{ \begin{bmatrix} 0 & +1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & +1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & +1 \\ 0 & -1 & 0 \end{bmatrix} \right\}$ ✗ $\rightarrow$ double neg is -1xans of (b)

29) a) All of $\mathbb{R}^3$ span space of all 3×3 matrices ✓

b) Line in $\mathbb{R}^3$ ✗ ~~does~~ (span space of all 3×3 matrices)

c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ All of $\mathbb{R}^3$ $\rightarrow$ each rank 1 matrix could contribute a unique column/matrix

span space of all 3×3 matrices ✗

$\rightarrow$ Span all multiples cI

31) a) All functions to satisfy $\frac{dy}{dx} = 0$.

$\rightarrow y(x) = c$

b) Choose particular function to satisfy $\frac{dy}{dx} = 3$:
 $y(x) = 3x$ ✓

c) Find all functions for $\frac{dy}{dx} = 3$:

$y(x) = 3x + c$ ✓
 $\rightarrow$??

$y = y_p + y_n$
 $\quad \quad \quad 3x \quad c$

34) $\dim = 1$: $y_1(x) = 3x$ $y_2(x) = 6(x)$ $y_3 = 9x$ ✓

$\dim = 2$: $y_1(x) = 3x$ $y_2(x) = x^2 + 7$ $y_3 = 6x$ ✗

$\dim = 3$: $y_1(x) = 3x$ $y_2(x) = x^3 + 7$ $y_3(x) = x^2$ ✗

$\rightarrow$ other examples : trig $\sin(x), \cos(x), \sin(2x)$
expon e^x, e^{2x}, e^{3x}